

RESEARCH PAPER

Cadmium and antibiotic-resistant *Acinetobacter calcoaceticus* strain STP14 reported from sewage treatment plant

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Abstract

A bacterium designated as strain STP14 was isolated from a sewage treatment plant and identified as *Acinetobacter calcoaceticus* based on 16S ribosomal RNA gene sequencing. Strain STP14 exhibited resistance to several metals such as mercury, cobalt, copper, nickel, lead, and cadmium. Among these metals, the bacterium showed maximum resistance to cadmium in concentration up to 1200 mg/L. The antimicrobial susceptibility test of *A. calcoaceticus* strain STP14 showed coresistance to all tested antibiotics except tigecycline and chloramphenicol for which 16 ± 1 - and 15 ± 1 -mm zone of inhibition was observed, respectively. The protein pattern of the crude cellular extract revealed substantial differences in protein bands of untreated control and cadmium treated *A. calcoaceticus* strain STP14 suggesting variable protein expression under cadmium stress. Metals and antibiotic resistance are increasing phenomenon and universal concern of public health. This study improves our understanding regarding the bacterial coresistance against metals and antibiotics and the possible emergence of multidrug resistance due to selective pressure and coselection in the metal polluted sewage sludge.

KEYWORDS

Acinetobacter calcoaceticus, antibiotic coresistance, cadmium, heavy metals, sewage

1 | INTRODUCTION

The rapid urbanization and industrialization have led to widespread metal pollution in environmental reservoirs including mine tailings and drainages, agricultural run-off, pharmaceutical, and industrial wastes as well as municipal wastewater [1]. Heavy metals are potentially toxic at relatively low concentrations and persist for a longer time, therefore, considered substantial environmental pollutants and concern to public health. Among the heavy metals, cadmium is a highly toxic and non-essential metal that is well recognized for its adverse

effects on the cellular system [2]. Cadmium-induced toxicity and carcinogenicity involve multiple organ damage-causing systemic toxicities even at a lower level of exposure [3].

The recurrent and tenacious pollution caused by toxic metals has a significant influence on the microbial community's composition and their ecological activities [4]. With its potential toxicity, the subinhibitory concentration of cadmium provides selective pressure towards bacteria helping them to adapt and develop resistance to confront environmental stress [5]. Antimicrobial resistance is one of the biggest threats to global

health accelerated by the inappropriate release of resistance-driving chemicals into the environment. The pool of antimicrobial-resistant bacteria and antimicrobial resistance genes in the environment is associated with contamination of soil and water bodies through untreated effluents and sewage [6,7]. Although the relative effect of cadmium on the prevalence of antibiotic resistance is not fully understood, it is reported that the coselection of cadmium can strongly affect bacterial abundance leading to metal and antibiotic coresistance [8]. For instance, cadmium resistance due to *czcR* provides a simultaneous resistance to cobalt, zinc, and coresistance to carbapenems by decreasing its influx to bacterial cells [9]. Similarly, resistance to cadmium has an evident role in the coselection of methicillin resistance in *Staphylococcus aureus* [10].

The metal coresistance is primarily important in which resistance genes to both metals and antibiotics are colocated together and can stimulate antibiotic resistance in the absence of antibiotics [11]. The long-term selective pressure caused by heavy metals due to their non-biodegradable and persistent nature can substantially influence horizontal gene transfer which in turn will propagate and proliferate antibiotic resistance, hence, forming a bacterial community resistant to antimicrobial agents [12]. The prevalence of such environmental bacteria with clinically important antibiotic resistance is an important component for the spread of resistant bacteria and in the progression of multidrug resistance [13].

Bacterial coresistance against heavy metals and antibiotics is a globally recognized threat to public health. Although antibiotic resistance has been mostly attributed to the upsurge in the use of antibiotics around the world

and little is known about the concurrent role of metals in antibiotic resistance, therefore, antibiotic resistance via selective pressure generated by metals needs to be addressed. The study aims to isolate cadmium-resistant bacteria from a sewage treatment plant (STP) and investigate its coresistance to antibiotics. This is the first-ever study reporting the occurrence of bacterial coresistance from the capital city of Pakistan.

2 | MATERIALS AND METHODS

2.1 | Chemicals and materials

The analytical grade reagents and chemicals were obtained from Sigma-Aldrich. All media used were purchased from OXOID, Thermo Scientific. Metal stocks were prepared and diluted in deionized water, fresh dilutions were used each time. Chemicals used for gel preparation were purchased from Bio-Rad. All laboratory glassware was cleaned by soaking in 2-M HNO_3 to remove any trace minerals and precipitates, rinsed with distilled water, and oven-dried for 2 h at 180°C.

2.2 | Sampling area and sample collection

The STP is located in the I-9 sector of Islamabad (33.650°N, 73.058°E; Figure 1). The plant receives municipal as well as hospital wastewater from 25 sectors. The total water treatment capacity of the plant is 17 million gallons per day [14]. The sludge samples were

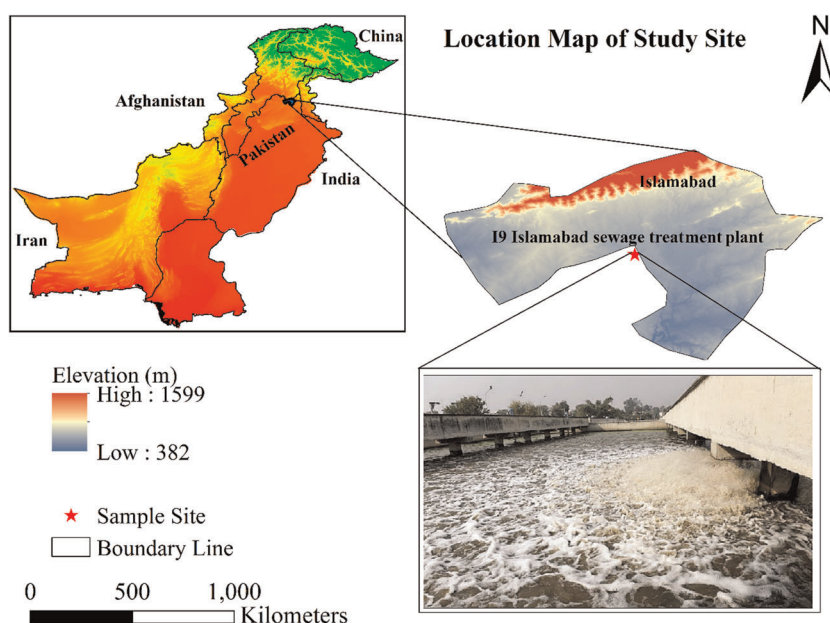


FIGURE 1 The sampling site of sewage treatment plant (STP) I-9, Islamabad, Pakistan

collected from STP in sterile, leak-proof Nalgene bottles by adopting standard sampling techniques [15], and dispatched to Applied Environmental and Geomicrobiology (AEG) laboratory, Quaid-i-Azam University Islamabad in a sealed box at 4°C and subsequently processed within 24 h for further experiments.

2.3 | Physicochemical analysis of samples

The physicochemical analysis of the sludge samples such as pH, temperature, moisture content, and metals, was carried out at the AEG laboratory. The pH of samples was noted by using Pt electrode with reference to an Ag/AgCl electrode (3.0-M KCl; BPP-922). The temperature of the study area was reported during sample collection and moisture content was measured according to the method described by Johnson [16]. Samples were prepared for metals analysis by adopting the method described by Din et al. [17]. Certified material National Institute of Standards and Technology 4355 was run parallel as a reference for analytical accuracy. Atomic absorption spectroscopy (AA-7000; Shimadzu) with an external calibration procedure was used for metals quantitative analysis. All the samples were analyzed in triplicate and mean values were calculated.

2.4 | Isolation of cadmium-resistant bacteria

Ten-gram sludge sample was inoculated in 90-ml nutrient broth and incubated overnight at 30°C and 120 rpm in a shaking incubator. The enriched bacterial culture was subsequently diluted 1:10 up to 10-folds and 0.1 ml of each dilution was spread on nutrient agar amended aseptically with 100 mg/L of cadmium chloride [18]. Plates were incubated at 30°C for 72 h and regularly checked for the appearance of bacterial colonies after every 24 h. Morphologically distinct bacterial colonies were picked up and purified by streaking on fresh medium supplemented with 100 mg/L of cadmium. This practice was performed several times and axenic cultures were kept on slants and stored at 4°C.

All isolates were screened for multiple metal tolerance by streaking on nutrient agar plates supplemented with chloride salts of individual metal at the following concentrations: 200–1400 mg/L for cadmium (Cd), cobalt (Co), copper (Cu), lead (Pb), and nickel (Ni) whereas 5–80 mg/L for mercury (Hg). Strain STP14 was selected

for further study based on tolerance to the highest concentration of cadmium along with other tested metals.

2.5 | Identification of strain STP14

Strain STP14 was grown overnight in 50 ml nutrient broth at 30°C and 120 rpm. The bacterial cells were harvested through centrifugation in the late exponential phase and genomic DNA was extracted by using a commercially available DNA extraction kit (Cat# 27300; Norgen Biotek Corp.) according to the manufacturer's instructions. The purity and quality of extracted DNA were analyzed by using NanoDrop (Fisher Scientific) and 260/280 was considered sufficient for polymerase chain reaction (PCR). The 16S ribosomal RNA (rRNA) gene was PCR amplified using universal primers 27F (5'-AGAGTTTGATCATGGCTCAG-3') and 1492R (5'-TACGGCTACCTTGTTACGACTT-3') [19]. PCR reaction mixtures contained 0.5 µl of each primer, 12.5 µl of DreamTaq master mix, 10.5 µl of deionized water, and 1-µl DNA template (10 ng/µl). The PCR conditions were: denaturation at 94°C for 2 min, 30 cycles of denaturation at 95°C for 1 min, annealing at 55°C for 1 min, extension at 72°C for 1 min, and a final extension step at 72°C for 10 min. PCR amplification was confirmed using a 1% agarose gel containing ethidium bromide (0.5 µg/ml) and analyzed using a Bio-Rad Gel Doc imaging system. The PCR amplicon was sent to Macrogen, Korea for 16S rRNA gene sequencing. The obtained sequence was BLAST searched (<http://www.ncbi.nlm.nih.gov/BLAST/>) in the nucleotide database of GenBank through BLASTn and closely related sequences were downloaded. All sequences were aligned, and a neighbor-joining phylogenetic tree was constructed using MEGA-X software [20]. The 16S rRNA nucleotide sequence was submitted to the National Center for Biotechnology Information under accession number MT733869.

2.6 | Characterization of strain STP14

2.6.1 | Determination of cadmium tolerance limit

A stock solution of 10,000 µg/ml of cadmium chloride was prepared. Fifty milliliters nutrient broth was supplemented with cadmium in concentration ranged from 200 to 1400 µg/ml in separate flasks followed by inoculation with 1% culture broth of strain STP14 grown overnight up to ~0.2 optical density (OD). All the flasks

were incubated at 30°C and 120 rpm in a shaking incubator and growth was measured (OD 600 nm) at 24-h interval through spectrophotometer (8453; Agilent).

2.6.2 | Optimum growth conditions

The optimum growth temperature and pH for strain STP14 were determined in 100 ml nutrient broth with 1% inoculum (OD ~ 0.2) and incubated at wide temperature (10–40°C) and pH (5.0–8.0) ranges in shaking incubator at 120 rpm. The growth was measured at OD 600 nm through spectrophotometer.

2.7 | Antibiotic susceptibility pattern of strain STP14

The antibiotic susceptibility pattern of strain STP14 was evaluated against 14 antibiotics (belonging to cell wall inhibitors, protein synthesis inhibitors, and DNA gyrase inhibitors) by the Kirby–Bauer disk diffusion method [21]. The selected antibiotics with disk content in µg were P: Penicillin (1 and 10), OX: oxacillin (1), FOX: cefoxitin (30), CTX: cefotaxime (30), CRO: ceftriaxone, VA: vancomycin (30), TGC: tigecycline (15), E: erythromycin (15), DA: clindamycin (2), CN: gentamicin (10), C: chloramphenicol (30), LZD: linezolid (30), and CIP: ciprofloxacin (5). The antibiotics discs were obtained from OXOID, Thermo Scientific. Strain STP14 was grown overnight (OD ~ 0.2) and a uniform bacterial lawn was made on freshly prepared Mueller Hinton agar followed by impregnation of antibiotics discs aseptically. The appropriate distance was maintained between the antibiotics disks and plates were incubated at 30°C for 24 h. After incubation, zones of inhibition were reported by measuring the diameter across the center of each zone in millimeters. Strain STP14 was considered as sensitive or resistant to the given antibiotic based upon the absence or presence of an inhibition zone following the Clinical and Laboratory Standards Institute recommended resistance breakpoints [22].

2.8 | Effect of cadmium on intracellular proteins

The effect of cadmium stress on the expression of intracellular protein was studied by adopting the method as previously described by Izrael-Zivkovic et al. [23]. Twenty-milliliter nutrient broth was taken in 100-ml Erlenmeyer flask supplemented with a sublethal concentration of CdCl₂ (200 µg/ml) and inoculated with 1% actively growing culture of strain STP14 (OD ~ 0.2). A cadmium-free control was run in parallel. The flasks were incubated in a shaking incubator at 30°C and 120 rpm for 14 h. Cells were harvested in the early stationary phase by centrifugation at 20,000g for 15 min and 4°C in 2-ml Eppendorf tubes. The pellet was dissolved in 100 µl of the 1X loading dye and heat shock was given for 5 min; the Eppendorf tube was shifted on ice for 2 min and subsequently centrifuged at 12,000g for 10 min. Proteins in the crude cellular extract were separated by electrophoresis on 12% sodium dodecyl sulfate–polyacrylamide gel (SDS-PAGE) with a 5% stacking gel using the mini-protean II system (Bio-Rad). The gel was run as described by Chovanova et al. [24] and a naked eye comparison of the cadmium stressed cells and untreated control was made to identify variation in protein bands due to cadmium exposure.

2.9 | Statistical analysis

All experiments were run in triplicates to calculate the mean ± standard deviation represented with error bars. A linear correlation analysis was conducted to determine whether there was a significant correlation between cadmium toxicity and tolerance. GraphPad Prism version 7.00 was used for data analysis and construction of graphs.

3 | RESULTS

3.1 | Sample analysis

Table 1 shows the physicochemical properties and elemental analysis of sludge samples collected from the STP.

TABLE 1 Physicochemical parameters and metals concentrations in the study sample

Sample	Physicochemical parameters			Metal concentration (mg/kg)				
	Temp (°C)	pH	Moisture (%)	Cu	Pb	Ni	Co	Cd
Study sample	30	6.5	80	84 ± 2	14 ± 3	47 ± 3	64 ± 3	24 ± 2
Control	30	7.2	5	2 ± 0.09	0.3 ± 0.06	3 ± 0.1	2 ± 0.1	0.3 ± 0.1

Note: Control = Garden Soil.

The temperature and pH were recorded as 30°C and 6.5, respectively. Upon metals analysis, the highest concentration was observed for copper (84 ± 2 mg/kg) while the least for lead (14 ± 3 mg/kg). Whereas cadmium was also detected in high concentrations (24 ± 2 mg/kg) in comparison to control.

3.2 | Isolation of cadmium-resistant bacterial strains

A total of 15 distinct bacterial strains designated as STP1 to STP15 were isolated. Table 2 presents a detailed view of bacterial tolerance to different metals at varying concentrations on nutrient agar plates and strain STP14 was found to be the most putative strain thus selected for further studies.

3.3 | Identification of strain STP14

The strain STP14 was grown overnight on nutrient agar at 30°C, and the colonies appeared smooth, mucoid, and greyish white with 1–2 mm in diameter. Gram-negative coccobacilli were observed upon microscopic examination. Figure 2 shows the phylogenetic tree, indicating the evolutionary relationship of strain STP14 to previously reported bacterial strains. The 16S rRNA sequence analysis of strain STP14 shows a close genetic relationship with *Acinetobacter calcoaceticus*.

3.4 | Characterization of strain STP14

Strain STP14 was characterized for its maximum tolerance limit against cadmium. A gradual decrease in optical density representing a decline in the bacterial growth ($R^2 = .9531$) indicating that strain STP14 can tolerate cadmium up to a concentration of 1200 µg/ml where no significant growth was observed above this limit (Figure 3). Strain STP14 can grow at a wide range of temperature (10–40°C) and pH (5.0–8.0) with optimum growth at 30°C and pH 7, respectively (Figure 4a,b).

3.5 | Antibiotic susceptibility pattern of *A. calcoaceticus* strain STP14

The antibiotic resistance pattern of metallotolerant *A. calcoaceticus* strain STP14 was studied against 14 antibiotics. Figure 5 presents the antibiogram of *A. calcoaceticus* strain STP14 indicating a significantly higher degree of resistance to 12 of 14 (86%) antibiotics tested.

Substantial growth inhibition of *A. calcoaceticus* strain STP14 was reported only in the case of tigecycline and chloramphenicol with an inhibition zone of 16 ± 1 and 15 ± 1 mm, respectively. Data on the antibiotic resistance pattern is summarized in Table 3.

3.6 | Intracellular protein profile

SDS-PAGE was performed to investigate differentially expressed protein bands of the intracellular proteins extracted from *A. calcoaceticus* strain STP14 grown in the presence and absence of cadmium. Figure 6 shows clear bands (marked from a to f) and a significant difference can be observed in protein bands of the test and control profiles. Interestingly, some protein bands (Figures 6; A, D–F) were overexpressed in the cadmium treated cells as compared to control, that is, without cadmium. Whereas some protein expression (Figure 6; B and C) was negatively affected by cadmium stress.

4 | DISCUSSION

The sewage sludge from wastewater contains nutrients, organic pollutants, and potentially toxic metals. Heavy metals are persistent pollutants that can adversely affect environmental health and have a significant contribution to the emergence of resistant microbial populations [25]. The STP of our study site receives municipal and hospital wastewater and holds higher concentrations of metals (Table 1), therefore, it provides a suitable niche for metallotolerant microorganisms. We isolated a total of 15 bacterial strain, *A. calcoaceticus* strain STP14 showed resistance to a higher concentration of all tested metals including cadmium and was further used in this study. Generally, bacterial communities are sensitive to the high concentration of heavy metals [26] that significantly decrease the bacterial biomass. However, a higher number of metallotolerant bacteria reported in this study shows the long-term polluted circumstances resulting in a selective pressure that consequently has significantly increased the metallotolerant bacterial diversity [27,28].

Acinetobacter spp. associated with multidrug resistance have been increasingly reported as a global threat to public health and environment, which is of concern [29]. Heavy metals form complex compounds that could be toxic to bacterial cells at higher concentrations. Therefore, microbial cells adopt defensive strategies to cope with long-term metal stresses [30]. Bacteria inhabiting metal-polluted environments are adapted to metal stress through various resistance mechanisms, primarily via metal efflux systems. The

TABLE 2 Tolerance profile of isolates against potentially toxic metals on nutrient agar medium

Isolates	Metals concentration (mg/L)														
	HgCl ₂					CoCl ₂					NiCl ₂				
	05	10	20	40	80	200	400	600	800	1000	200	400	600	800	1000
STP1	R	-	-	-	-	R	-	-	-	-	R	-	R	R	-
STP2	R	-	-	-	-	-	-	-	-	-	-	-	-	-	-
STP3	R	R	R	R	-	R	-	-	-	-	R	-	R	R	-
STP4	R	-	-	-	-	-	-	-	-	-	-	-	-	-	-
STP5	R	-	-	-	-	-	-	-	-	-	-	-	-	-	-
STP6	R	-	-	-	-	R	-	-	-	-	R	-	-	-	-
STP7	R	-	-	-	-	-	-	-	-	-	-	-	-	-	-
STP8	R	-	-	-	-	-	-	-	-	-	-	-	-	-	-
STP9	R	-	-	-	-	R	R	-	-	-	R	-	R	R	-
STP10	R	R	R	-	-	R	R	-	-	-	R	R	R	R	-
STP11	R	R	R	-	-	R	R	-	-	-	R	R	R	R	-
STP12	R	-	-	-	-	R	-	-	-	-	R	-	-	-	-
STP13	R	R	R	R	-	R	-	-	-	-	R	R	R	R	-
STP14	R	R	R	R	-	R	R	R	R	-	R	R	R	R	R
STP15	R	R	R	-	-	R	R	R	R	-	R	R	R	R	-

Abbreviations: R, resistant (tolerated); -, sensitive (inhibited).

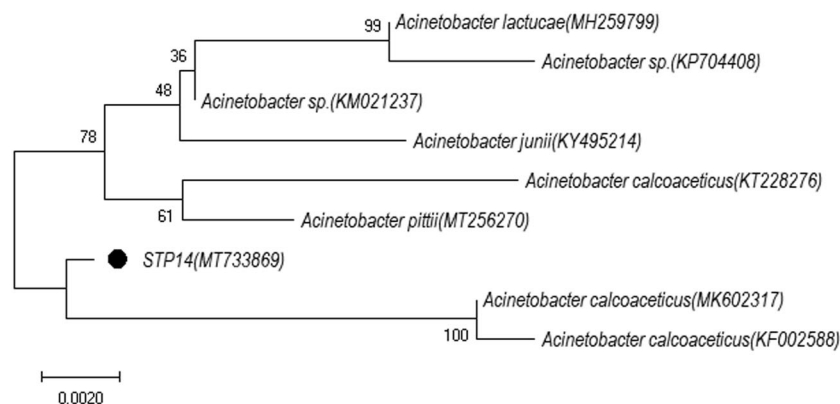


FIGURE 2 Phylogenetic tree of strain STP14 constructed by the neighbor-joining method indicates its close evolutionary relationship to *Acinetobacter calcoaceticus*. The bootstrap value was 1000 replicates and are reported as percentages. The scale bar corresponds to 0.0020 change per nucleotide

optimum growth of *A. calcoaceticus* strain STP14 at 30°C and neutral pH is correlated to the physicochemical properties of the sampling site [31]. This is the pioneer study where *A. calcoaceticus* coresistant to multimetals and antibiotics is reported in sewage sludge. *A. calcoaceticus* is an opportunistic pathogen and causes nosocomial infections worldwide [32]. Therefore, the existence of *A. calcoaceticus* in this environment could be directly linked with the hospital's wastewater in the STP.

The existence of multiple stresses such as heavy metals, chemical reagents, disinfectants, and antibiotics (generally higher in-hospital wastewater) could provide a favorable condition for bacterial resistance to all these stresses. *A. calcoaceticus* strain STP14 exhibited resistance to maximum antibiotics used in the current study. The reduced antimicrobial susceptibility could be due to its exposure to heavy metals in sewage sludge contributing to the coselection of antibiotic resistance. The findings of Garhwal et al. [33] that metal exposure significantly changes antibiotic resistance index and the suggested mechanism of bacterial coresistance against heavy metals and antibiotics at the genetic level by

Gillings et al. [34] supports our results. According to Nguyen et al. [35], cadmium and zinc are the most common heavy metals associated with bacterial resistance against antibiotics. Similarly, Pal et al. [36] reported a novel connection between zinc/cadmium and aminoglycoside/macrolide resistance genes. It is widely accepted that metal tolerant bacteria confer co-resistance to a variety of antibiotics by expression of resistant genes under metal stress. Kaur et al. [37] suggested a strong correlation between metal and antibiotic resistance by exposing *Salmonella enterica* serovar Typhi to cadmium

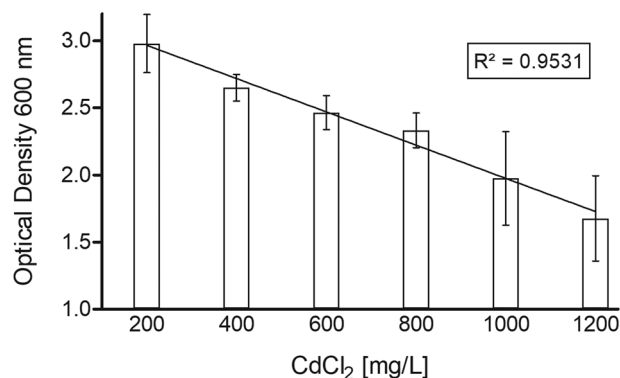


FIGURE 3 Tolerance limit of bacterial strain STP14 against a range of CdCl_2 concentrations (200–1200 mg/L) with linear correlation ($R^2 = .9531$) after 24 h of incubation. Error bars represent standard deviation ($n = 3$)

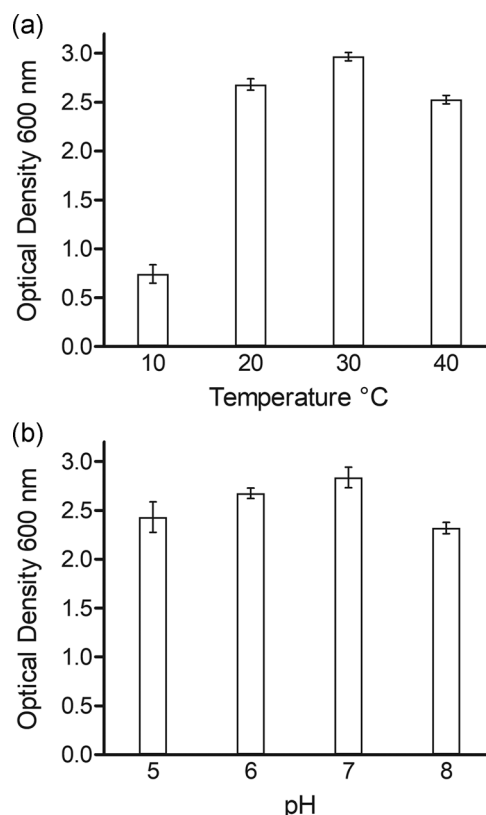


FIGURE 4 Optimization of growth temperature (a) and pH (b) for bacterial strain STP14. Error bars represent $\pm$ SD; ($n = 3$)

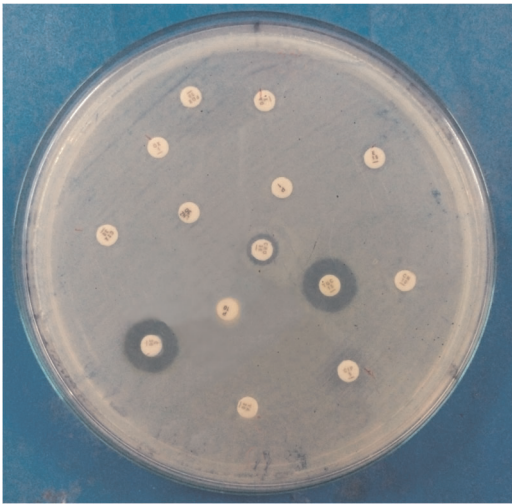


FIGURE 5 Antibiotic resistance assay of *Acinetobacter calcoaceticus* strain STP14. The clear inhibition zone is shown around tigecycline and chloramphenicol

that consequently resulted in antibiotic resistance after adaptation.

The cadmium and zinc-resistance in bacteria are due to the two main efflux systems mediated by the RND-driven (resistance-nodulation-cell division protein family) transporter system and P-type ATPase efflux system [38]. The heavy metal tolerance mechanism that involves protein synthesis leads to variation in the protein profile of tolerant bacteria. The same was observed when *A. calcoaceticus* strain STP14 was grown under metal stress (Figure 6). These proteins can be ribosomal protein (translation of natural messenger RNAs), or protein for cell growth and division. It may be heat shock protein in line with previous studies showing the cadmium-induced upregulation of heat shock proteins. Overexpression of the heat-shock proteins in the cell treated with cadmium may be critical for incapacitating variation that contains protein denaturation brought by cadmium [39]. The coresistance of bacterial strain STP14 against antibiotics and cadmium strongly suggests that this resistance is genetically related. According to Knapp et al. [40], a correlation exists between metals and antibiotic resistance genes. Chen et al. [28] reported a significant correlation between cadmium and target resistance genes. Metal repossession may cause increased antibiotic resistance and it may work coregulatory with antibiotic stress. A variety of gene expression and protein responses to these stresses may be linked to their co-ordination. In addition to the long-term persistence in a higher concentration of heavy metals in the studied sludge, the hospital wastewater could also provide favorable conditions for antibiotic resistance to *A. calcoaceticus* strain STP14. The genes on a plasmid of bacteria

TABLE 3 Resistance/sensitivity pattern of the bacterial *Acinetobacter calcoaceticus* strain STP14 to different antibiotics

Tested antibiotic	Penicillin P1	Penicillin P10	Oxacillin OX1	Cefoxitin FOX30	Cefotaxime CTX30	Ceftriaxone CRO30	Vanco mycin VA30	Tigecycline TGC15	Erythro mycin EI5	Clinda mycin DA2	Genta micin CN10	Chloramphe nicol C30	Line zolid LN230	Cipro floxacin CIP5
Zone of inhibition (mm)	0	0	0	0	0	9 ± 0.5	0	16 ± 1	0	0	0	15 ± 1	0	0
Susceptibility pattern	R	R	R	R	R	R	R	S	R	R	R	S	R	R

Abbreviations: R, resistant; S, sensitive.

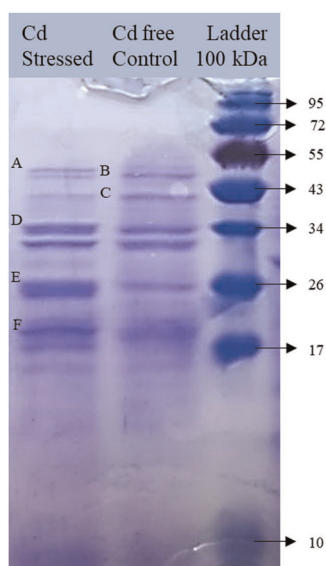


FIGURE 6 Polyacrylamide gel electrophoresis (12%) of the proteins extracted from the cadmium-resistant *Acinetobacter calcoaceticus* strain STP14 grown in the presence or the absence of cadmium. Protein bands showing variations are labeled through A–F

responsible for co-resistance against heavy metals and antibiotics proposes the exertion of stress implemented by the contamination of metals and antibiotics in their environment [41]. Yamina et al. [42] also reported cadmium-resistant bacteria from hospital wastewaters and found the strains resistant to antibiotics.

The presence of such metallotolerant and antibiotic-resistant opportunistic pathogens in environmental samples is a huge concern for society. The possibility of horizontal gene transfer, especially the antimicrobial resistance genes do exist in the contemporary environmental microorganisms [43], the previous study provided potential evidence of horizontal gene transfer (plasmids mediated and integrons based) in wastewaters habitats [44]. This emerging and increasing risk of health needs proper and serious attention. Generally, sludge is used as fertilizer that might enhance the risk of spread of these metals and antibiotics coresistant opportunistic pathogens into the environment. The sludge should be checked for the presence of any sort of lethal ingredients including heavy metals, antibiotics, and pathogenic microbes before using it as a fertilizer. Therefore, a proper and safe subsequent processing of the sludge after wastewater treatment is crucial for the authorities. Moreover, under selective pressure, human pathogens can become antibiotic-resistant and STP provides a suitable environment for this phenomenon. Therefore, more systematic studies including statistical data that can describe the response relationship of bacteria towards heavy metals

and antibiotics exposure in STP should be conducted. Besides, molecular-based culture-independent studies are recommended to get a better insight into bacterial co- and cross-resistance mechanisms to antibiotics and heavy metals.

In the current study, antibiotics, and metals coresistant bacteria *A. calcoaceticus* strain STP14 was isolated from a STP. It was also observed that the protein profile of strain STP14 was affected by cadmium stress. This study improves our understanding regarding the bacterial coresistance against metals and antibiotics as well as highlights the importance of metal pollution in STPs that could boost the emergence of multidrug resistance via coselection.

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CONFLICT OF INTERESTS

The authors declare that there are no conflict of interests.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author.

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